Deepak Manoharan

Avionics System Verification & Validation Engineer

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Professional Summary

Avionics verification and validation engineer with 3.5 years of experience in avionics software development and requirements-based, system-level testing of safety-critical embedded systems. Currently develops and executes DOORS-derived test scripts on mini-bench setups for Touch Screen Controller modules on the Honeywell G300 program, covering ASCB databus communication and cockpit display behavior against DO-178B requirements. Earlier work includes 2,000+ test cases for 550+ requirements on the Spanish F-5 program, executed in simulation and on target hardware, and interface and display software for the Airbus Helicopters Helionix platform. Strong working knowledge of system interfaces, system architecture, ARINC-429 and ARINC-653 databuses, requirements traceability, and DO-178B/C and MIL-STD processes.

Core Skills

- **Verification & Validation:** Black-box (requirements-based) testing, test script development and Python test automation, High-Level Testing (HLT), Low-Level Testing (LLT), MC/DC coverage, test design and planning, mini-bench and simulation bench execution, hardware-in-the-loop (HIL) testing, system integration testing, cockpit display and touchscreen verification, test results and defect analysis, anomaly reproduction and root-cause identification
- **Requirements & Traceability:** Requirements analysis, interface requirements, signal-to-display mapping, requirements traceability (DOORS, Marcel Impact, custom traceability automation tool), certification support
- **System Interfaces & Databuses:** ARINC-429, ARINC-653 (IMA), ASCB, RS-485, I2C, Ethernet, discrete I/O, analog signal interfaces (RPM, EGT, OILP)
- **Test & Simulation Tools:** TATS, TIU, ESCAPE, EASE, MMT, TestProcess, WinTest, Mystic, Excaliber, WinView, LCDM, RTRT, CANTATA
- **Development & Configuration Tools:** GILDA, KOJAK, Visual C++ 2015 (licensed), Eclipse, Git, Jenkins, PCF, IRS
- **Standards & Process:** DO-178B/C; MIL-STD-130, MIL-STD-461D, MIL-STD-704A, MIL-STD-1686A; MIL-L-85762A; V-Model; Agile
- **Programming:** C, C++, Python, Java

Professional Experience

Cyient Limited | Mar 2025 – Present

Verification and Validation Engineer

Project: Honeywell G300 aircraft program

- Develop test scripts from requirements in DOORS, building test logic per MC/DC coverage criteria and executing them on mini-bench setups.
- Execute requirements-based testing for Touch Screen Controller (TSC) avionics modules, including TGF, TFM, TIO, TCF, UDPTSC, UDPAGM and INAV.
- Verify cockpit display pages and touchscreen functionality against DO-178B software requirements.
- Simulate and validate TINK/ASCB databus communication using TestProcess and customized TFM executables on mini-bench setups.
- Use Python for test automation, defect analysis and execution of test scripts derived from DOORS requirements.
- Use simulation tools including TATS, TIU, ESCAPE, EASE and MMT for test execution and display validation.

Airbus India (via Expleo Solutions) | Feb 2025 – Jan 2026

Software and System Engineer

Project: Helionix (Airbus Helicopters), a modular Integrated Modular Avionics (IMA) glass-cockpit avionics suite

Developed cockpit interface and display software on an Integrated Modular Avionics (ARINC-653) architecture, mapping aircraft sensor and equipment signals to the MFDs and the Aircraft Management Computer (AMC) and enabling simulation, target deployment and DO-178 compliant deliverables.

- Authored interface requirements and implemented signal-to-display mappings from system architecture diagrams to MFD/AMC inputs.
- Delivered signal files and outputs used to validate runtime behavior on the simulator and on target hardware.
- Supported system integration and certification by maintaining requirements traceability in DOORS and resolving interface issues with software and systems teams.
- Integrated signal artifacts into Git/Jenkins build pipelines, working in line with DO-178C objectives.

- Created and maintained signal definition files in GILDA and used KOJAK to auto-generate embedded C source code for simulation, target-based development and R&D validation.

Expleo Solutions, India | Jun 2023 – Feb 2025

Software Engineer (Apr 2024 – Feb 2025) | **Associate Software Engineer** (Jun 2023 – Feb 2024)

Project: Electronic Engine Display (EED) and Multi-Functional Display (MFD) for Spanish F-5 aircraft

Developed and verified the EED and MFD for real-time visualization of engine, fuel and navigation data, with ARINC-429 integration and Built-In Test (BIT) support, in compliance with military avionics standards.

- Analyzed 550+ requirements and developed 2,000+ test cases and test scripts to verify them, executed in both simulation and on target hardware for the Spanish F-5.
- Conducted real-time ARINC-429 testing using Mystic and WinTest and verified system behavior across all operational modes, using a cross-compiler in Visual C++ 2015.
- Collaborated on instrumentation setup and ground simulation to reproduce and analyze in-service anomalies, supporting effective root-cause identification.
- Participated in flight test support, including real-time system behavior analysis and defect reporting based on flight test outcomes.
- Developed a traceability automation tool integrating the requirement lifecycle across four key documents, significantly reducing manual tracking effort.
- Authored Software Design Documents (SDD) and Software Verification Cases & Procedures (SVCP).
- Implemented core EED display software in C++ to compute and render real-time engine parameters (RPM, Nozzle Position, EGT, FF, FQ, OP), including data filtering, unit conversion and display formatting.

Education

Bachelor of Engineering, Government College of Engineering (Affiliated to Anna University), Salem, Tamil Nadu, India | Graduated 2023 | CGPA: 8.7

Certifications & Awards

- ISTQB Certified Tester – Foundation Level
- Applause Award, Rising Star